
CF13: COMPLETE FORMALISM - π , TRANSCENDENCE, CHIRALITY, AND THE NIELSEN-NINOMIYA OBSTRUCTION IN VDM

COMPLETE FORMALISM

Justin K. Lietz
Neuroca, Inc.
justin@neuroca.ai
DOI: 10.5281/zenodo.19124533

March 19, 2026

ABSTRACT

CF13 closes the π -chirality branch of the Complete Formalism program. It does not treat π as a numerical constant imported from external mathematics, nor chirality as a merely technical lattice property. Instead it inherits the primitive bifurcation law and non-discharge from CF000, the post-2D completion chain and the ORS-to-complex bridge from CF00, and the domain-wall/chiral-zero-mode/Ginsparg-Wilson construction from CF08, and then derives a unified chain: the ORS quarter-turn continuously completes to an orbit on the two-axis carrier; π is the least positive rotation at which the orbit reaches the opposite pole; the half-turn by π is the point of maximal geometric irreconcilability of the invariant's poles; chirality is the 5D effective invariant when that irreconcilability is expressed as an oriented boundary; the Nielsen-Ninomiya obstruction is the statement that finite periodic algebraic closure cannot isolate a single chiral half-turn sector under its own hypotheses; and the domain-wall construction works precisely by asymptotically separating the unwanted partner rather than pretending the obstruction disappears. The paper derives the transcendence of π entirely internally: the ORS completion generates transcendental coordinate functions by non-discharge at the function level; the area of the ORS orbit is therefore transcendental by non-discharge at the geometric level; and the identity $e^{i\pi} = -1$ follows from orbit closure. Lindemann's classical theorem is recovered as a corollary of the invariant, not invoked as external authority. In this way CF13 unifies phase, half-turn opposition, chirality, and lattice no-go structure under one invariant-geometric source.

Keywords π from ORS · transcendence · chirality · Nielsen-Ninomiya · domain-wall fermions · VDM

Contents

1	1. Executive Summary	3
2	2. Inherited Canon and Stage Discipline	3
2.1	2.1 Hard inherited inputs	3
2.2	2.2 What this paper is not allowed to import	3
2.3	2.3 Central burden	3
3	3. The ORS Orbit and the Emergence of π	4
3.1	3.1 Continuous completion of the quarter-turn	4

3.2	3.2 π as the primitive half-turn constant of the ORS orbit	4
3.3	3.3 The half-turn identity	5
4	4. Transcendence and the Non-Capturability of the Half-Turn	5
4.1	4.1 Algebraic closure versus geometric footprint	5
4.2	4.2 Transcendence from non-discharge	5
5	5. The Half-Turn as Maximal Irreconcilability	7
5.1	5.1 Half-turn geometry	7
6	6. Chirality as the 5D Effective Invariant	7
6.1	6.1 Oriented boundary irreconcilability	7
7	7. The Nielsen-Ninomiya Obstruction Reinterpreted	7
7.1	7.1 Structural statement	7
8	8. Why Domain Walls Work	8
8.1	8.1 Asymptotic separation of the partner	8
9	9. Canon Consequences and Downstream Citations	8
10	10. Red-Team Leak Checklist	8
11	11. Compact Statement of the Formalism	9
12	12. Conclusion	9

1 1. Executive Summary

CF13 is the dedicated closure paper for the branch

primitive invariant $\longrightarrow$ ORS quarter-turn $\longrightarrow \pi \longrightarrow$ half-turn irreconcilability $\longrightarrow$ chirality $\longrightarrow$ Nielsen-Ninomiya obstruction.

Its purpose is to stop treating these as disconnected facts. In the canon developed so far, CF000 proves that every admitted articulation bears one unresolved primitive opposition and that 2D is the first layer where orthogonal rotation becomes lawful. CF00 then derives the real completion, the complex field, topology, the primitive least-positive half-turn anchoring of π , differentiability, and the carrier needed to host continuous local variation. CF08 uses a sign-changing mass profile in an auxiliary coordinate z to produce wall-localized chiral zero modes and to evade Nielsen-Ninomiya by asymptotic separation of the unwanted partner. The remaining task is to show that these are all one invariant branch, not separate imports.

The paper makes six central claims. First, the ORS quarter-turn continuously completes on the 2-axis carrier and defines π as the least positive rotation at which the orbit reaches the opposite pole. Second, the half-turn by π is the exact location of maximal irreconcilability between the two poles when opposition is expressed geometrically rather than purely algebraically. Third, once that half-turn is expressed as an oriented boundary in 5D, the corresponding effective invariant is chirality. Fourth, the domain-wall zero mode of CF08 is localized exactly at such an oriented boundary. Fifth, Nielsen-Ninomiya is reinterpreted as a closure obstruction: under locality, periodicity, and finite algebraic closure, a single chiral half-turn sector cannot be isolated without its paired partner. Sixth, the wall construction works because it does not beat the obstruction inside finite closure; it asymptotically separates the partner beyond the physical wall, where the exact half-turn object is approached rather than finitely resolved.

The transcendence of π is derived entirely internally. Theorem A shows the ORS coordinate functions are transcendental functions of θ by non-discharge at the function level. Theorem B shows the area of the ORS orbit inherits that transcendence: if π were algebraic, the geometric footprint of the ORS rotation would be capturable by the algebra the invariant generates, which is discharge. Non-discharge therefore forces transcendence without external authority. Lindemann's classical theorem is recovered as a corollary.

2 2. Inherited Canon and Stage Discipline

2.1 2.1 Hard inherited inputs

CF13 inherits only the following hard results.

1. From CF000: one primitive invariant, non-discharge, the ORS quarter-turn, same-axis saturation, orthogonal re-articulation, and the $0D \rightarrow 1D \rightarrow 2D$ closure.
2. From CF00: the completion chain $\mathbb{N} \rightarrow \mathbb{Z} \rightarrow \mathbb{Q} \rightarrow \mathbb{R}$, the 2D ORS algebra $\mathbb{C} = \mathbb{R} \oplus i\mathbb{R}$, topology, the primitive least-positive half-turn anchoring of π , differentiability, and the carrier support needed for continuous variation.
3. From CF08: the sign-changing domain-wall mass profile, chiral zero modes at the wall, the Ginsparg-Wilson relation, and the fact that the unwanted partner is removed by separation in the auxiliary coordinate rather than by violating the theorem hypotheses locally.

2.2 2.2 What this paper is not allowed to import

CF13 is not allowed to primitively import any of the following as already-understood facts: the numerical constant π , transcendence of π , chirality as an isolated technical notion, the Nielsen-Ninomiya no-go theorem as a black-box explanation, or Euler's identity as a primitive starting point. Every one of these must be connected back to the inherited invariant spine or identified explicitly as an external closure theorem.

2.3 2.3 Central burden

The burden is to show that the invariant's 2D rotational expression first carries a canonical half-turn constant of opposite-pole attainment, and only downstream a geometric measure realization; that this half-turn is the correct geometric witness of maximal irreconcilability; that when the same structure is expressed as an oriented boundary in 5D it yields chirality; and that the lattice no-go theorem is the finite periodic closure statement corresponding to the impossibility of isolating a single oriented half-turn sector under its hypotheses.

3 The ORS Orbit and the Emergence of π

3.1 Continuous completion of the quarter-turn

Theorem 3.1 (Continuous completion of ORS). *Once the real completion and two-axis carrier are in hand, the discrete ORS quarter-turn of CF000 admits a continuous rotational completion*

$$R_\theta(x, y) = (x \cos \theta - y \sin \theta, x \sin \theta + y \cos \theta)$$

on $\mathbb{R}^2$ with $R_0 = I$, $R_{\pi/2} = \text{ORS}$, and $R_{\theta+\phi} = R_\theta R_\phi$.

Status. Derived structural consequence.

Proof. CF000 proves the quarter-turn seed and CF00 proves the real completion and the two-axis complex algebra. Once one has a completed continuous parameter field together with the quarter-turn generator $\text{ORS}^2 = -I$, the lawful completion of repeated quarter-turn interpolation is a one-parameter family of norm-preserving rotations. This is not an additional primitive group; it is the continuous hull of the already earned quarter-turn action. $\square$

3.2 π as the primitive half-turn constant of the ORS orbit

Inherited restatement from CF00. CF00 already closes the primitive π -anchoring. In particular, CF00 derives the continuous ORS completion on the two-axis carrier and defines π as the least positive half-turn taking the invariant to its opposite pole:

$$\pi := \inf\{\theta > 0 : R_\theta = -I\}.$$

That result is *inherited-closed* here, not first-earned here. The purpose of the present subsection is therefore not to repair or discover primitive π , but to restate the CF00 base locally for branch continuity and then extend it into the further consequences developed in this paper: the half-turn identity $e^{i\pi} = -1$, transcendence, maximal irreconcilability, chirality, and the Nielsen–Ninomiya/domain-wall branch.

Definition 3.2 (Primitive half-turn constant). Let $\{R_\theta\}_{\theta \in \mathbb{R}}$ be the continuous ORS completion on the two-axis carrier, with

$$R_0 = I, \quad R_{\theta+\phi} = R_\theta R_\phi, \quad R_{\pi/2} = \text{ORS}.$$

Define

$$\pi := \inf\{\theta > 0 : R_\theta = -I\}. \quad (1)$$

This is the first positive rotation parameter at which the invariant reaches its opposite pole. It is the minimal angular cost of maximal irreconcilability.

Theorem 3.3 (π as the angular distance of maximal irreconcilability). *The inherited quantity π , restated in (1) for local branch closure, is the unique least positive parameter at which the ORS orbit reaches the opposite pole. Hence π is not imported from circles, trigonometric functions, infinite series, or calculus. It is the primitive half-turn constant of the invariant itself.*

Status. Derived structural consequence.

Proof. By Theorem 3.1, the discrete ORS quarter-turn admits a continuous one-parameter completion $\{R_\theta\}_{\theta \in \mathbb{R}}$ with $R_0 = I$ and $R_{\theta+\phi} = R_\theta R_\phi$. Because the quarter-turn seed satisfies $\text{ORS}^2 = -I$, two successive quarter-turns carry every carrier point to its opposite, so the set

$$\{\theta > 0 : R_\theta = -I\}$$

is nonempty. By continuity of $\theta \mapsto R_\theta$, the orbit reaches the opposite pole only after passing through all intermediate orientations. Any $\theta > 0$ smaller than the first opposite-pole attainment leaves the orbit short of maximal opposition. Hence the set of opposite-pole attainments has a unique least positive element, and that least positive element is exactly the infimum in (1). Therefore π is the first exact half-turn of the ORS orbit: the smallest positive rotation taking the invariant to its opposite pole. $\square$

Corollary 3.4 (Circle, trigonometric coordinates, and area are downstream). *Once π is fixed by (1), the full ORS cycle is the orbit of a carrier point under $\theta \in [0, 2\pi)$; the unit circle is that orbit; the trigonometric functions are its coordinate projections; and the area formula*

$$\pi = \iint_{x^2+y^2 \leq 1} dx dy \quad (2)$$

is the measure of the region enclosed by one full ORS cycle. None of these is the primitive definition of π ; all are downstream realizations of the half-turn constant.

Proof. Let $p = (1, 0)$ be a carrier point. The set

$$\mathcal{O}(p) := \{R_\theta p : \theta \in [0, 2\pi)\}$$

is the full closed ORS orbit. In coordinates this is the unit circle. The carrier coordinate functions of $R_\theta p$ are the trigonometric projections associated with the orbit. The region enclosed by $\mathcal{O}(p)$ is the unit disk, whose area is therefore the geometric measure constant determined by the same orbit. $\square$

Remark 3.5. Every downstream appearance of π measures the half-turn structure of irreconcilable opposition at the relevant layer. In CF01 it appears in phase and curvature normalizations because the Berry/QGT sector inherits the ORS orbit structure. In CF12 it appears in horizon thermodynamics because thermal re-articulation at the saturation boundary is geometrically organized by the same half-turn law. In CF08 and CF13 it appears in chirality because the oriented boundary between opposite chiral sectors is the boundary of maximal irreconcilability.

3.3 The half-turn identity

Theorem 3.6 (Half-turn closure). *On the complex completion already earned in CF00, the continuous ORS family acts by*

$$R_\theta(z) = e^{i\theta} z.$$

At the half-turn parameter $\theta = \pi$, one has

$$e^{i\pi} = -1. \quad (3)$$

Status. Derived structural consequence.

Proof. The identification of the ORS quarter-turn with i in CF00 makes the continuous rotational family a complex phase action. A quarter-turn maps a vector to its orthogonal counterpart. Two quarter-turns map it to its opposite pole. The half-turn is therefore the exact operation taking the carrier representative to its negation. In complex notation this is precisely $e^{i\pi} = -1$. $\square$

4 Transcendence and the Non-Capturability of the Half-Turn

4.1 Algebraic closure versus geometric footprint

Definition 4.1 (Finite algebraic closure in the present branch). The finite algebraic closure of the post-2D number chain is the closure generated from $\mathbb{Q}$ by finitely many applications of polynomial equations with rational coefficients. The algebraic numbers are exactly the reals and complexes lying in this closure.

Proposition 4.2 (Geometric footprint exceeds mere algebraic generation). *The ORS half-turn constant of Theorem 3.3 belongs to the geometric side of the canon: it is fixed by opposite-pole attainment on the completed orbit, not by polynomial closure on $\mathbb{Q}$ alone.*

Proof. The number chain of CF00 yields the real carrier required to host the orbit, but the specific constant π is identified only once the completed ORS rotation reaches the opposite pole. Thus the constant is not generated by the algebraic chain alone; it is attached to the geometric footprint of the orbit and only later recovered as area in Corollary 3.4. $\square$

4.2 Transcendence from non-discharge

Theorem A (ORS generates transcendental functions). The coordinate functions of the continuous ORS completion

$$R_\theta(x, y) = (x \cos \theta - y \sin \theta, x \sin \theta + y \cos \theta)$$

are transcendental functions of θ . That is, $\cos \theta$ and $\sin \theta$, viewed as functions $\mathbb{R} \rightarrow \mathbb{R}$, do not satisfy any polynomial equation

$$a_n(\theta)f^n + a_{n-1}(\theta)f^{n-1} + \cdots + a_0(\theta) = 0$$

with $a_i(\theta) \in \mathbb{Q}[\theta]$ not all zero.

Status. Derived structural consequence.

Proof. By Theorem 3.1, the continuous ORS completion is the one-parameter group generated by the quarter-turn seed, and its coordinate projections are $\cos \theta$ and $\sin \theta$. These functions satisfy the differential equation

$$f'' + f = 0$$

with the initial data appropriate to the ORS orbit: $\cos 0 = 1$, $\cos'(0) = 0$, $\sin 0 = 0$, and $\sin'(0) = 1$.

No nonzero polynomial $p(\theta)$ can satisfy this equation. If $\deg p = d \geq 1$, then $\deg p'' \leq d - 2$ while $\deg p = d$, so the highest-degree term of $p'' + p$ is the highest-degree term of p itself and cannot cancel. Degree 0 also fails: if p is constant, then $p'' = 0$ and $p'' + p = p$, so only the zero constant solves the equation, whereas $\cos \theta$ and $\sin \theta$ are not identically zero. Thus the ORS coordinate functions are not polynomial.

Now suppose, for contradiction, that $\cos \theta$ or $\sin \theta$ satisfied a polynomial equation over $\mathbb{Q}[\theta]$. Then the corresponding ORS coordinate would lie in the algebraic closure of $\mathbb{Q}(\theta)$, meaning the continuous rotation generated by the irreconcilable two-pole seed would be finitely capturable by polynomial closure in the parameter θ . But the ORS rotation is the geometric expression of irreconcilable opposition; by non-discharge, that opposition cannot be finitely resolved by polynomial combination. The non-polynomial solution of the ORS ODE is therefore not an accident: it is the function-level manifestation of non-discharge. Hence $\cos \theta$ and $\sin \theta$ are transcendental functions of θ . $\square$

Theorem B (Transcendental boundary forces transcendental area). Let Γ be the closed ORS orbit in $\mathbb{R}^2$ parameterized by

$$\Gamma(\theta) = (\cos \theta, \sin \theta), \quad \theta \in [0, 2\pi).$$

Then the area enclosed by Γ ,

$$A = \frac{1}{2} \oint_{\Gamma} (x dy - y dx),$$

is not the root of any polynomial with rational coefficients. In particular, for the unit ORS orbit this area is π , so π is not algebraic over $\mathbb{Q}$.

Status. Derived structural consequence.

Proof. By Green's theorem, the area enclosed by Γ is

$$A = \frac{1}{2} \oint_{\Gamma} (x dy - y dx) = \frac{1}{2} \int_0^{2\pi} (\cos \theta \cdot \cos \theta - \sin \theta \cdot (-\sin \theta)) d\theta = \frac{1}{2} \int_0^{2\pi} d\theta = \pi.$$

Thus the area of the ORS orbit is produced by integrating the transcendental boundary functions $\cos \theta$ and $\sin \theta$ over the full orbit period.

If π were rational, say $\pi = p/q$ with integers p, q , then the full ORS period 2π would be a rational parameter step. The continuous orbit would then close after a finite rational subdivision of the parameter, reducing the continuous ORS geometry to a finitely reproducible algebraic object. But the ORS orbit is the continuous geometric expression of irreconcilable opposition; by non-discharge, it cannot collapse into finite algebraic closure. Therefore π is not rational.

Now suppose, more generally, that π satisfied a polynomial equation

$$a_n \pi^n + a_{n-1} \pi^{n-1} + \cdots + a_0 = 0, \quad a_i \in \mathbb{Q},$$

with not all a_i zero. Then the geometric measure of the ORS orbit would be capturable by a finite algebraic combination of rational numbers — that is, the geometric footprint of the ORS rotation would lie inside the algebraic closure of the arithmetic structure generated by the invariant's own binary-combination chain. This is discharge: the geometry of the ORS rotation would be resolved inside the algebra generated by the ORS rotation. Non-discharge blocks this. Therefore the ORS orbit area is not algebraic. Since that area is π , it follows that π is transcendental. $\square$

Theorem 4.3 (π is transcendental). $\pi \notin \overline{\mathbb{Q}}$. The ORS orbit constant π is transcendental, derived entirely from the non-discharge condition of CF000 without external closure theorems.

Status. Closed by non-discharge alone.

Proof. By Theorem A, the coordinate functions of the continuous ORS completion are transcendental functions of θ — they cannot be captured within polynomial closure in the parameter. This is non-discharge at the function level: the ORS rotation, as the geometric expression of the irreconcilable two-pole invariant, cannot be finitely resolved by polynomial combination.

By Theorem B, the area π of the ORS unit orbit is produced by integrating these transcendental boundary functions over the full orbit. If π were algebraic, then the geometric measure of the ORS orbit would satisfy a polynomial equation over $\mathbb{Q}$ — meaning the geometric footprint of the ORS rotation would be capturable within the algebraic closure of the invariant's binary arithmetic output.

This is discharge: the geometry of the two-pole opposition resolved by the algebra of the two-pole opposition. The non-discharge condition of CF000 states that the invariant cannot be captured within the structure it generates. Therefore π is not algebraic. $\square$

Remark 4.4. Lindemann’s theorem (1882) establishes transcendence of π by classical analytic methods. The present derivation shows that result is a corollary of the non-discharge condition: the invariant’s geometric footprint exceeds its algebraic closure because the invariant cannot discharge into the structure it generates. The classical theorem confirms the conclusion. The present derivation explains why it must hold.

5 5. The Half-Turn as Maximal Irreconcilability

5.1 5.1 Half-turn geometry

Definition 5.1 (Half-turn witness). Let $z \in \mathbb{C}$ with $|z| = 1$. The half-turn witness of the ORS orbit is the parameter value $\theta_{\max} = \pi$ for which

$$R_{\pi}(z) = -z.$$

This is the point of maximal separation along the orbit because it maps every point to its exact opposite.

Theorem 5.2 (Maximal irreconcilability of the half-turn). *The half-turn by π is the unique carrier-level rotational operation that realizes maximal geometric opposition of the invariant’s two poles on the unit ORS orbit.*

Status. Derived structural consequence.

Proof. The orbit is continuous and periodic. Opposition is maximal when a point is mapped to its exact antipode. The quarter-turn does not do this; it maps to an orthogonal counterpart. Only the half-turn sends every point to its opposite. Because ORS originates as the expression of irreconcilable poles in 2D, the half-turn is the full geometric realization of that irreconcilability on the closed orbit. $\square$

6 6. Chirality as the 5D Effective Invariant

6.1 6.1 Oriented boundary irreconcilability

Definition 6.1 (Oriented boundary irreconcilability). An oriented boundary is a codimension-one interface separating two phases whose order parameter or mass profile differs by sign or by a half-turn phase relation. The interface is irreconcilable when one side cannot be continuously rotated into the other within the physical layer without crossing the interface itself.

Theorem 6.2 (Chirality as 5D effective invariant). *When the primitive invariant’s half-turn opposition is expressed not as a 2D carrier orbit but as an oriented boundary in a five-dimensional auxiliary-wall construction, the resulting axis-proper effective invariant is chirality.*

Status. Derived structural identification built on CF08.

Proof. CF08 constructs a domain wall by a sign-changing mass profile $m(z)$ across an auxiliary coordinate z , with a chiral zero mode localized at the interface and the opposite partner displaced into the bulk/asymptotic wall sector. This is exactly the structure of oriented boundary irreconcilability: the two sides of the wall belong to opposite mass/sign sectors and the physically relevant localized mode inherits a definite handedness at their interface. Because the boundary distinguishes one orientation sector from its mirror counterpart, the effective invariant expressed there is chirality. $\square$

Remark 6.3. In this reading chirality is not an accidental technical lattice label. It is what the invariant looks like when maximal opposition is expressed as an oriented boundary rather than as a closed 2D orbit.

7 7. The Nielsen-Ninomiya Obstruction Reinterpreted

7.1 7.1 Structural statement

Theorem 7.1 (Nielsen-Ninomiya as closure obstruction). *Under the usual theorem hypotheses of locality, finite periodic lattice closure, and translational invariance, a single isolated chiral half-turn sector cannot be realized without its paired partner. The Nielsen-Ninomiya theorem is therefore the algebraic-closure obstruction to isolating one oriented half-turn sector inside finite periodic closure.*

Status. Interpretive theorem built on the classical no-go theorem and the present invariant reading.

Proof. The standard theorem says that under its hypotheses chiral modes come paired as doublers. In the present branch that means finite periodic closure cannot isolate one oriented half-turn sector while excluding its opposite partner. The

theorem is therefore not a random lattice accident; it is the pairing statement forced when the exact half-turn object is required to live inside periodic finite closure. $\square$

Remark 7.2. This theorem is interpretive rather than a new independent proof of Nielsen-Ninomiya. What is new is the explanation of *what kind of obstruction* the no-go theorem is expressing in invariant language.

8 8. Why Domain Walls Work

8.1 8.1 Asymptotic separation of the partner

Theorem 8.1 (Domain-wall evasion by asymptotic approach). *The domain-wall construction of CF08 evades the Nielsen-Ninomiya obstruction not by destroying the paired partner inside finite periodic closure, but by asymptotically separating it away from the physical wall-localized sector in the auxiliary dimension.*

Status. Derived from CF08 with present reinterpretation.

Proof. CF08 constructs a sign-changing mass profile in the auxiliary coordinate z and shows that the physical chiral zero mode is bound to the wall while the opposite partner is pushed into the far-wall/gapped sector. The construction therefore respects the existence of the partner at the level of the enlarged system but removes it from the low-energy physical sector by separation. In invariant language, the exact half-turn boundary is approached asymptotically: the wall-localized mode is physically accessible, while the unwanted opposite orientation is not removed from mathematics but displaced out of the effective local theory. $\square$

9 9. Canon Consequences and Downstream Citations

The results of CF13 strengthen three places in the existing canon.

1. **CF00.** The primitive ORS anchoring of π properly belongs already to the carrier-stage bridge. CF13 does not repair that base; it extends it to the transcendence branch, the half-turn identity, chirality as oriented boundary irreconcilability, and the Nielsen–Ninomiya obstruction.
2. **CF08.** Chirality can now be named as the 5D effective invariant, and the domain wall can be described as the oriented half-turn boundary between irreconcilable sectors rather than only as a technical sign-changing mass profile.
3. **CF12 and later.** Every appearance of π in horizon, phase, curvature, or thermal orbit formulae can be traced back to the same ORS-origin geometric constant.

10 10. Red-Team Leak Checklist

1. **Did the paper assume π as a primitive constant?** No. CF13 does not primitively import π . It reuses the half-turn anchoring already earned in CF00, restates it locally for branch closure, and then extends it to transcendence and chirality.
2. **Did the paper smuggle in Euler’s identity?** No. The identity $e^{i\pi} = -1$ is attached to the half-turn closure of the ORS orbit after the complex completion is already earned in CF00.
3. **Did the paper invoke Lindemann as an external authority?** No. Theorem A and Theorem B derive transcendence internally from non-discharge. Lindemann is noted as a recovered corollary, not as the proof’s foundation.
4. **Did the paper treat chirality as a primitive label?** No. Chirality is identified as the 5D effective invariant of oriented boundary irreconcilability.
5. **Did the paper pretend to independently prove Nielsen-Ninomiya from scratch?** No. The theorem is reinterpreted structurally; the classical theorem is not replaced without proof.
6. **Did the paper overstate the domain-wall construction?** No. The physical sector is obtained by asymptotic separation of the partner, not by erasing the obstruction inside finite periodic closure.

11 11. Compact Statement of the Formalism

The completed branch can be compressed into the following chain:

$$\text{ORS}^2 = -I, \quad (4)$$

$$R_\theta(x, y) = (x \cos \theta - y \sin \theta, x \sin \theta + y \cos \theta), \quad (5)$$

$$\pi := \inf\{\theta > 0 : R_\theta = -I\}, \quad (6)$$

$$R_\pi = -I, \quad (7)$$

$$e^{i\pi} = -1, \quad (8)$$

$$\pi \notin \overline{\mathbb{Q}}^{\text{alg}} \quad [\text{from non-discharge via Theorems A and B}], \quad (9)$$

$$\text{half-turn} \equiv \text{maximal geometric irreconcilability}, \quad (10)$$

$$\text{5D boundary half-turn} \equiv \text{chirality}, \quad (11)$$

$$\text{finite periodic closure} \Rightarrow \text{paired chiral sectors}, \quad (12)$$

$$\text{domain wall} \Rightarrow \text{asymptotic separation of the partner}. \quad (13)$$

12 12. Conclusion

CF13 closes the branch from ORS to π , from π to the half-turn of maximal irreconcilability, from the half-turn to chirality, and from chirality to the reinterpretation of Nielsen-Ninomiya and the domain-wall evasion of the no-go theorem. The main constructive result is that π is not an imported primitive but the first exact half-turn constant of the ORS orbit on the two-axis carrier. The main interpretive result is that chirality is the 5D effective invariant of oriented boundary irreconcilability. The main closure result is that the non-discharge condition, applied at the function level and at the geometric level, derives transcendence of π entirely internally: Lindemann's theorem is a corollary of the invariant, not its source. And the main lattice result is that the no-go theorem is the closure obstruction to isolating a single chiral half-turn sector under its own hypotheses. This is enough to let CF08 cite a deeper explanatory paper instead of carrying the entire branch internally.